

Capstone Project Presentation

Integrated Space Weather Monitoring and Satellite Operations Decision Support System

CSA0911- Programming in Java



Candidates

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Introduction

- Space weather events like **solar flares and geomagnetic storms** can affect satellite operations.
- These events may cause problems in **communication, navigation, and satellite systems**.
- This project monitors and analyzes **space weather conditions in real time**.
- It helps predict possible **risks and impacts on satellites**.
- The system provides **decision support and operational recommendations** to improve satellite safety and reliability.

Abstract

Brief summary of objective and key results

- The **Integrated Space Weather Monitoring and Satellite Operations Decision Support System** is designed to monitor space weather conditions and assess their impact on satellite operations.
- The system collects and analyzes data related to **solar flares, geomagnetic storms, solar wind, and other space weather events**.
- It identifies potential risks that may affect **satellite communication, navigation, power systems, and overall performance**.
- Based on the analyzed data, the system provides **alerts, risk levels, and operational recommendations**.
- The main goal is to support **faster decision-making, improve satellite safety, and reduce the impact of space weather events**.

Objectives

- Monitor space weather conditions such as solar flares, solar wind, and geomagnetic storms.
- Collect and analyze space weather data to identify potential risks.
- Predict the impact of space weather events on satellite operations.
- Generate alerts for critical or dangerous space weather conditions.
- Provide decision support and operational recommendations for satellite management.
- Improve satellite safety, reliability, and operational efficiency.

Problem Statement

Derived from literature review and research gaps

- Space weather events such as solar flares, solar storms, and geomagnetic disturbances can negatively affect satellite operations.
- These events may cause communication failures, navigation errors, power disruptions, and satellite damage.
- Existing monitoring systems may not provide integrated risk analysis and timely operational recommendations.
- Therefore, there is a need for a system that can monitor space weather, analyze satellite risks, generate alerts, and support operational decision-making.
- The proposed system aims to provide early warnings and intelligent recommendations to improve satellite safety and reliability.

Literature Review / Background

Journals referred for research work

Year	Research / Author	Key Findings
2024	Kasapis & Kitiashvili (NASA)	AI/ML can improve prediction of solar activity and Solar Energetic Particle events.
2024	Eastwood et al.	L5-based monitoring can provide earlier warnings of hazardous space-weather events.
2024	Montagna et al.	Space-weather forecasts can support satellite operations and improve mission performance.
2025	Shirke & Parihar	AI models were used to predict solar flares, CME travel time, geomagnetic storms, and satellite risks.
2026	SURROUND Mission Concept	Advanced spacecraft monitoring can improve detection and tracking of CMEs and Solar Energetic Particles.

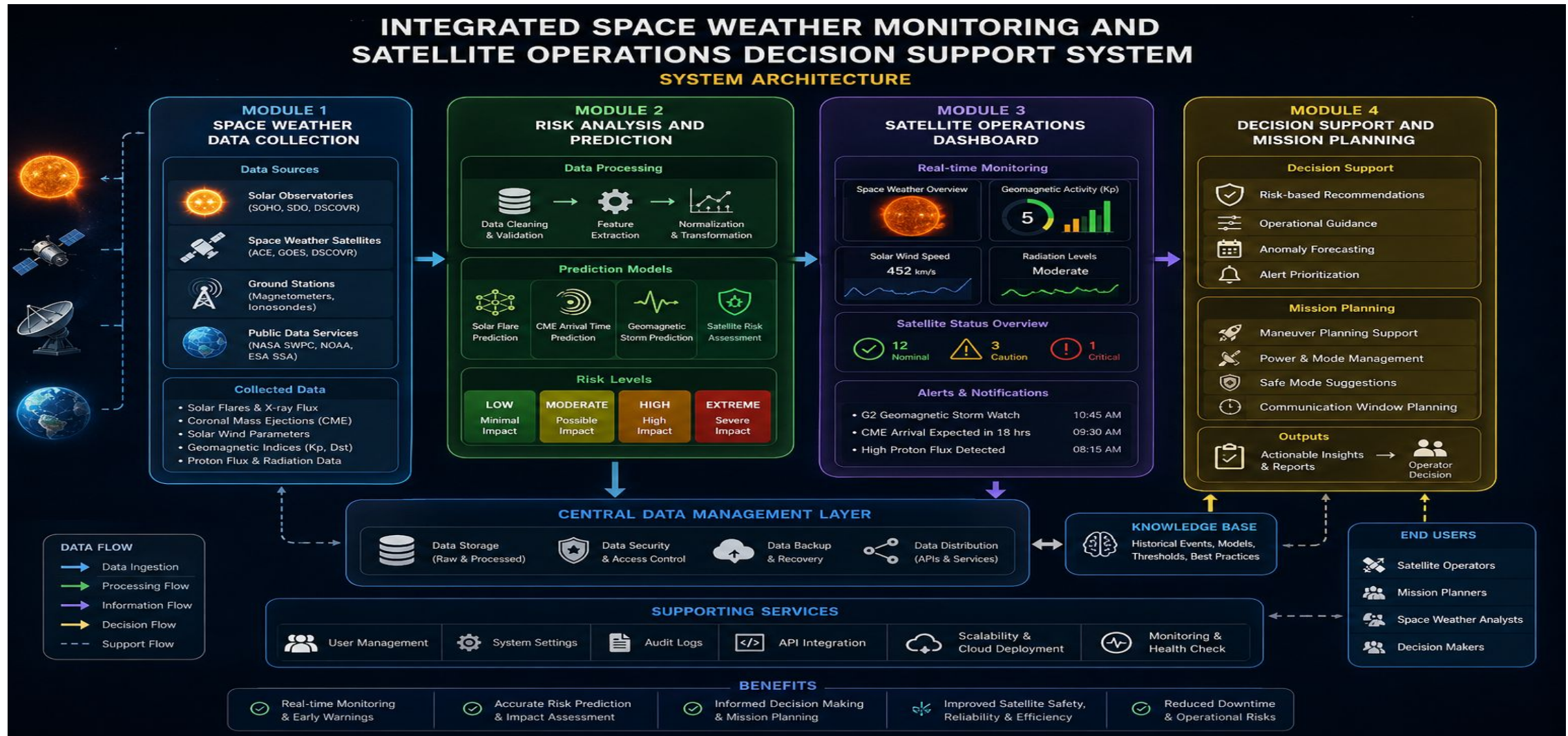
Summary

Recent studies show increasing use of **AI, real-time monitoring, and space-weather forecasting** to protect satellite operations. However, most existing systems focus on individual prediction or monitoring tasks. **Our project combines monitoring, risk assessment, alerts, and operational recommendations into one integrated decision-support system.**

System Architecture / Workflow

Show complete flow across modules

Architecture



Module 1: Space Weather Data Collection

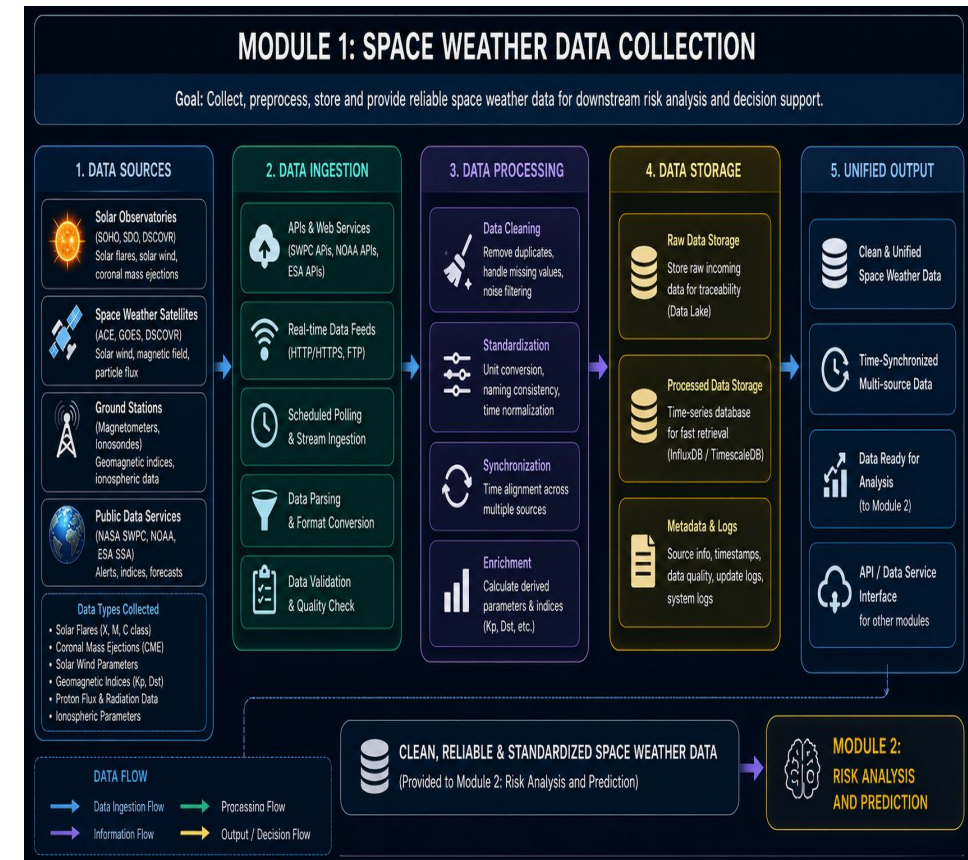
Approach, tools, techniques, and justification

High-Level Approach

- **Collect Data** – Gather space-weather data from satellites, solar observatories, ground stations
- **Real-Time Ingestion** – Receive data continuously through APIs, web services, and live data feeds.
- **Data Processing** – Parse, standardize, synchronize, and validate the incoming data.
- **Data Storage** – Store raw and processed data in a structured database for historical and real-time analysis.
- **Unified Output** – Provide clean and reliable space-weather data to **Module 2: Risk Analysis and Prediction**.

Justification

- Space weather data comes from **multiple heterogeneous sources**, so integration is required.
- **Real-time collection** helps detect hazardous events early.
- Data validation improves the **accuracy and reliability** of predictions.
- Centralized storage enables **quick access and historical analysis**.
- This module provides the **reliable data foundation** required for the remaining modules.



Module 2: Risk Analysis and Prediction

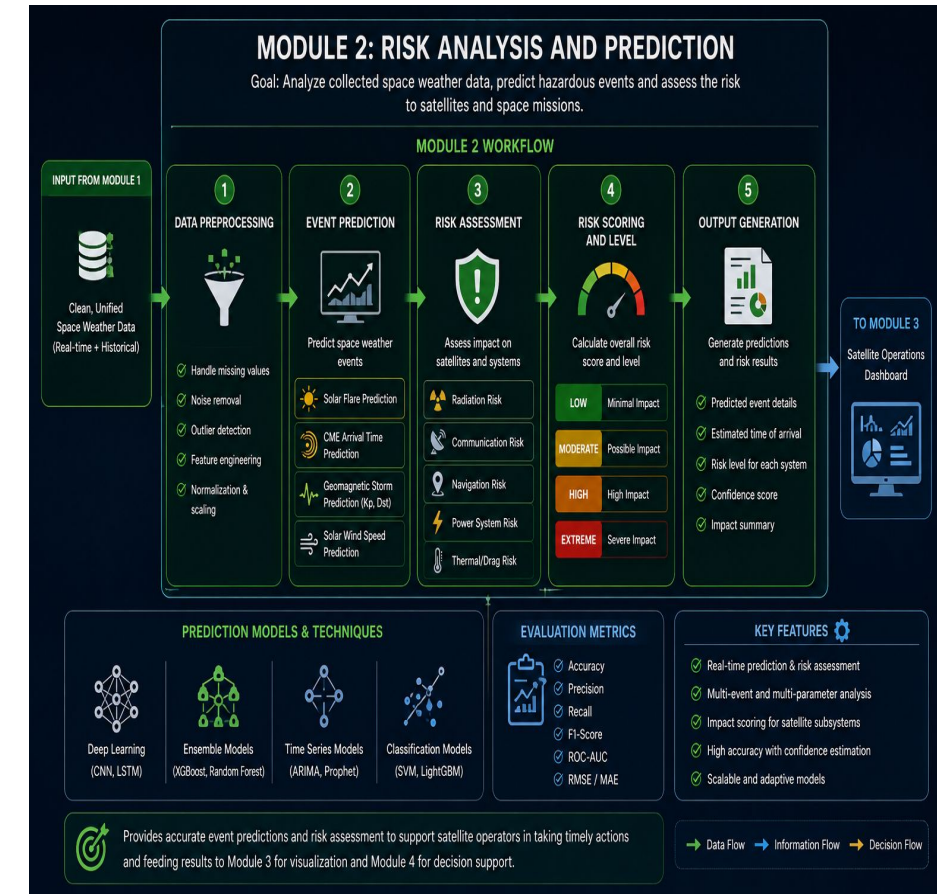
Approach, tools, techniques, and justification

High-Level Approach

- **Data Preprocessing** – Clean and prepare the space-weather data received from Module 1.
- **Event Prediction** – Predict events such as **solar flares, CMEs, and geomagnetic storms**.
- **Impact Assessment** – Analyze how predicted events may affect satellite communication, navigation, power, and other subsystems.
- **Risk Scoring** – Calculate a risk score and classify the threat as **Low, Moderate, High, or Extreme**.
- **Result Generation** – Generate predicted events, arrival time, risk level, confidence, and impact information for the next modules.

Justification

- Early prediction helps satellite operators **prepare for hazardous space-weather events**.
- Risk scoring converts complex space-weather data into **easy-to-understand threat levels**.
- Impact assessment helps identify **which satellite systems are most vulnerable**.
- Combining multiple prediction models can improve **forecasting reliability**.
- The results provide the foundation for **real-time monitoring in Module 3 and decision-making in Module 4**.



Module 3: Satellite Operations Dashboard

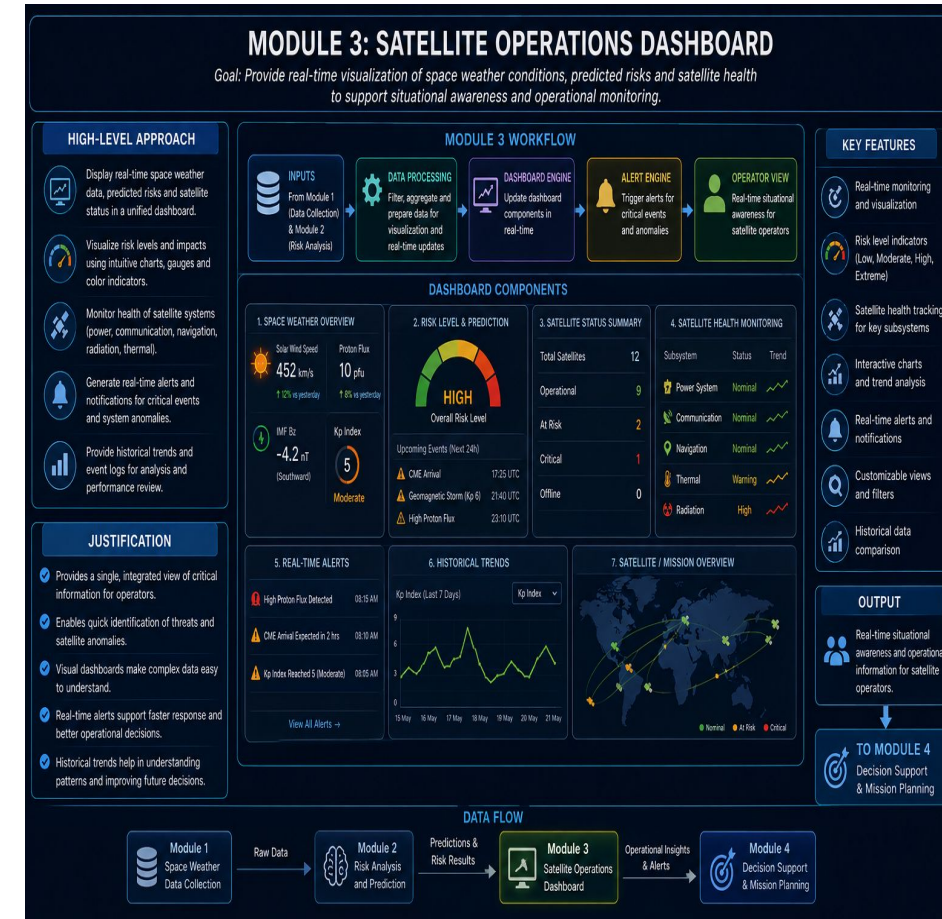
Approach, tools, techniques, and justification

High-Level Approach

- **Real-Time Monitoring** – Display live space-weather conditions and satellite status in one dashboard.
- **Risk Visualization** – Show predicted risks from Module 2 using clear **Low, Moderate, High, and Extreme** levels.
- **Satellite Health Monitoring** – Track important parameters such as **power, communication, navigation, radiation, and thermal conditions**.
- **Alerts & Notifications** – Generate immediate alerts when critical space-weather conditions or satellite risks are detected.
- **Historical Analysis** – Provide charts and trends to compare current conditions with previous events.

Justification

- Provides a **single view** of space-weather and satellite conditions.
- Helps operators identify **critical situations quickly**.
- Visual risk levels make complex data **easy to understand**.
- Real-time alerts support **faster response to hazardous events**.
- Historical trends help operators understand **patterns and previous incidents**.
- Connects prediction results with **practical satellite operations and decision-making**.



Module 4 : DECISION SUPPORT AND MISSION PLANNING

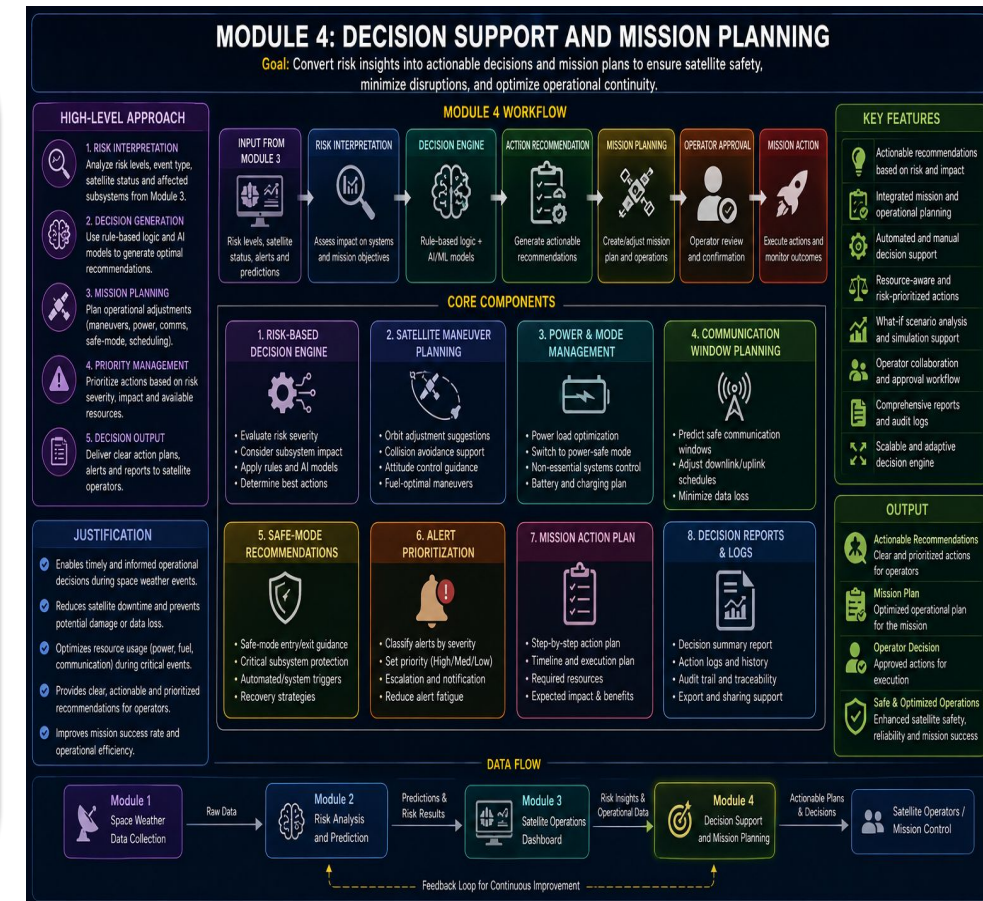
Approach, tools, techniques, and justification

High-Level Approach

- **Risk Interpretation** – Analyze the risk levels and satellite conditions received from Module 3.
- **Decision Generation** – Convert risk information into clear and actionable operational recommendations.
- **Mission Planning** – Support decisions related to satellite maneuvers, communication windows, power management, and safe-mode operations.
- **Priority Management** – Prioritize critical alerts and recommend appropriate actions based on risk severity.
- **Decision Output** – Provide recommendations, alerts, and mission-planning information to satellite operators.

Justification

- Helps operators take **quick and informed decisions** during space-weather events.
- Converts complex predictions into **simple operational actions**.
- Reduces the possibility of **satellite damage and mission disruption**.
- Supports **safe-mode activation, power management, and communication planning**.
- Improves overall **mission safety, reliability, and operational efficiency**.



Build

Software Functionality

Input

Processing

Output

Result And Discussion

- The system successfully **collects and integrates space-weather data** from multiple sources.
- The risk analysis module identifies events such as **solar flares, CMEs, and geomagnetic storms** and estimates their potential impact on satellites.
- The dashboard provides **real-time visualization of space-weather conditions, satellite health, risk levels, and alerts**.
- The decision-support module generates **risk-based recommendations** for power management, communication, safe mode, and mission planning.
- Overall, the integrated system improves **situational awareness, response time, satellite safety, and operational decision-making**.
- The results demonstrate that combining **monitoring, prediction, visualization, and decision support** can provide a more effective approach to managing space-weather-related satellite risks.

Conclusion & Future Work

Achievements and scope for improvement

Achievements

- Successfully integrated **multi-source space-weather data collection**.
- Developed **risk analysis and prediction** for major space-weather events.
- Created a **real-time satellite operations dashboard**.
- Implemented **risk-based alerts and operational recommendations**.
- Established an integrated workflow from **data collection to mission decision-making**.

Limitations

- Prediction accuracy depends on the **quality and availability of input data**.
- Real-time data may experience **API delays or connectivity issues**.
- Prediction models may require **large historical datasets** for better accuracy.
- The system provides decision support, but **final mission decisions require human/operator validation**.

Future Work

- Integrate **advanced AI and deep-learning models** for improved prediction accuracy.
- Add **live satellite telemetry** for more accurate health and risk assessment.
- Develop **automated anomaly detection and adaptive risk prediction**.
- Support **digital-twin and simulation-based mission planning**.

References

Use IEEE / APA / Springer format

Journal and Conference Publications

- **Eastwood, J. P., et al. (2024).** *The Vigil Magnetometer for Operational Space Weather Services From the Sun–Earth L5 Point.* **Space Weather, 22.**
- **Kasapis, S., & Kitiashvili, I. (2024).** *Forecasting Space Weather Using Machine Learning.* **NASA Technical Reports Server / SC24.**
- **Shirke, R. R., & Parihar, V. (2025).** *AI-Based Space Weather Prediction for Satellite Protection.* **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, X-5/W2, 585–595.**
- **Yu, et al. (2026).** *Solar Energetic Particle Forecasting With Multi-Task Deep Learning: SEPNET.* **Journal of Geophysical Research: Machine Learning and Computation.**
- **Khalimonenko, N., Kozak, L., Petrenko, B., & Kronberg, E. A. (2026).** *Forecasting Geomagnetic Kp Index Using Long Short-Term Memory-Attention Model.* **Journal of Atmospheric and Solar-Terrestrial Physics, 284, 106839.**



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